

Zhao Shizhen

Computer Science | Quantitative Finance | AI/Data Solutions |
Backend Systems | Client-Facing Technical Delivery

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EDUCATION

National University of Singapore - Bachelor's Degree in Computer Science, Double Major in Quantitative Finance 2025 - 2029 | Singapore
GPA: 4.85/5.0 | **Achievements:** Dean's List AY25/26 Semester 2; University Project Showcase 2026 Top 2 Winner
Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Discrete Structures, Linear Algebra, Calculus II

WORK EXPERIENCE

Kpler - Forward Deployed Engineer Intern, Commodities May 2026 - Present

- Designed data solutions over commodity and maritime datasets, translating client needs into technical deliverables.
- Built Python/SQL backend tools for historical extractions, data processing, and delivery.
- Partnered with Product and Sales through the end-to-end sales cycle, owning technical scoping and customized demos.

Source Academy - DevOps ML Engineer, Internal Learning Platform Jan 2026 - May 2026

- Built an LLM chatbot with an internal knowledge base and Retrieval-Augmented Generation (RAG) techniques.
- Deployed models on AWS Lambda and Docker for scalable inference.
- Improved inference speed and accuracy by over 50% through efficient chunking methods.

Shopee - Product Manager Intern, Internal HR Systems Sep 2025 - Dec 2025

- Designed agentic AI workflows using prompt engineering and tool integration to streamline HR processes.
- Integrated LLMs with APIs, databases, and internal systems, reducing processing time by 90%.
- Coordinated with the development team to schedule requests within delivery pipelines while balancing user needs and engineering bandwidth.

PROJECTS

O(Alpha) - Quant Research & Agentic Trading Platform github.com/shiverin/O-Alpha

- Built a Go/Next.js platform with TimescaleDB, Redis, Docker, and dashboards for portfolio research and paper trading.
- Implemented validation and paper-trading loops with walk-forward testing, cost stress checks, and portfolio snapshots.

Messenger - Real-Time Full-Stack Messaging App github.com/shiverin/messenger

- Built a private chat app with Django Channels, WebSockets, PostgreSQL, a custom SPA-style UI, secure authentication, and persistent storage.

Traffic Sign Classification with CNN github.com/shiverin/CNNTraffic

- Achieved 98.97% accuracy on GTSRB by designing and optimizing TensorFlow CNN architectures.
- Compared convolutional depth, pooling, dense capacity, and dropout to improve generalization.

TECHNICAL SKILLS

Languages: Python, SQL, Go, TypeScript, JavaScript, Java, C, R
Backend/API: REST APIs, Gin, FastAPI, Django, Flask, WebSockets, PostgreSQL/TimescaleDB, Redis
Data/Analytics: Snowflake, AWS Athena, ETL/data workflows, Pandas, Scikit-learn, Tableau, R Studio
AI/ML: LLMs, RAG, TensorFlow, PyTorch, YOLOv8, Whisper
Cloud/Dev Tools: AWS Lambda, Docker, Docker Compose, Git, Postman, Render, Supabase

HACKATHONS

BrainHack by DSTA - Finalist (Top 4) Jun 2025

- Fine-tuned Whisper on 4,500 audio samples, achieving 0.98% word error rate.
- Built a ResNet18 + CRNN model on 6,000 scanned text samples, achieving 99% character-level accuracy.
- Trained YOLOv8 on 10,000 labelled military-vehicle images, achieving 67% test accuracy.
- Implemented a single-policy multi-agent DQN in a custom Scout-vs-Guards environment using PettingZoo.

Interests: Classical musician (piano and violin), orchestral performance, Leetcode, watch collecting, Chinese chess